

Townsend 2.5-2.8

Questions for Gallicchio

I don't understand why the identity operator is introduced, not the difference between using intermediary states a_1 and a_2 that result from the use of the identity operator. Aka, explain this quote:

"

When we do not make a measurement that permits us to distinguish the intermediate states $|a_1\rangle$ and $|a_2\rangle$, we add the amplitudes and then square to get the probability, while if

we do make a measurement that can distinguish which of the states $|a_1\rangle$ and $|a_2\rangle$ the particle is in, we add the individual probabilities, not the amplitudes.

"

2.5 Changing Representations

We can use the rotation operator to change to a new basis by rotating the $|+z\rangle$ that we normally see to a $|+x\rangle$

$$R^{\dagger}\left(\frac{\pi}{2}\vec{j}\right)|+z\rangle = |+x\rangle$$

and can do this on any state $|+n\rangle$ with any phase around any axis.

Passive rotation: The basis is changed, but we have the same overall thing

Active rotation: We change the thing

S basis:

$$\hat{A} \xrightarrow{S_x} S^{\dagger} \hat{A} S$$

I.e.,

$$\begin{aligned} S &= \begin{pmatrix} \langle +z|+x\rangle & \langle +z|-x\rangle \\ \langle -z|+x\rangle & \langle -z|-x\rangle \end{pmatrix} \\ &= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \end{aligned}$$

2.6 Expectation values

$$\langle S_z \rangle = \left(\frac{\hbar}{2}\right) |\langle +z|\psi \rangle|^2 + \left(-\frac{\hbar}{2}\right) |\langle -z|\psi \rangle|^2$$

$$|+x\rangle = \frac{1}{\sqrt{2}}|+z\rangle + \frac{1}{\sqrt{2}}|-z\rangle \quad |-x\rangle = \frac{1}{\sqrt{2}}|+z\rangle - \frac{1}{\sqrt{2}}|-z\rangle$$

TION In Example 2.5 we saw that in this basis

$$\hat{J}_z \rightarrow \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$\langle S_z \rangle = (1, 0) \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 0$$

2.7 Photon Polarization

infinite small angle rotations take a pure x to a pure y photon.

Light acts like a spin $\frac{1}{2}$ particle.

Basis in circular coordinates - right or left hand because easy eigen vectors. Then change basis with matrices.

EXAMPLE 2.8 Determine the matrix representation of the angular momentum operator $\hat{J}_z$ using both the circular polarization vectors $|R\rangle$ and $|L\rangle$ and the linear polarization vectors $|x\rangle$ and $|y\rangle$ as a basis.

SOLUTION Let's start with the easy one first. Since the states $|R\rangle$ and $|L\rangle$ are eigenstates of $\hat{J}_z$ with eigenvalues $\hbar$ and $-\hbar$, respectively

$$\hat{J}_z \xrightarrow{|R\rangle-|L\rangle \text{ basis}} \begin{pmatrix} \langle R|\hat{J}_z|R\rangle & \langle R|\hat{J}_z|L\rangle \\ \langle L|\hat{J}_z|R\rangle & \langle L|\hat{J}_z|L\rangle \end{pmatrix} = \begin{pmatrix} \hbar & 0 \\ 0 & -\hbar \end{pmatrix}$$

The matrix is diagonal in this basis with the eigenvalues of the basis states on the diagonal. Switching to the linear polarization states $|x\rangle$ and $|y\rangle$:

$$\begin{aligned} \hat{J}_z \xrightarrow{|x\rangle-|y\rangle \text{ basis}} & \begin{pmatrix} \langle x|R\rangle & \langle x|L\rangle \\ \langle y|R\rangle & \langle y|L\rangle \end{pmatrix} \begin{pmatrix} \langle R|\hat{J}_z|R\rangle & \langle R|\hat{J}_z|L\rangle \\ \langle L|\hat{J}_z|R\rangle & \langle L|\hat{J}_z|L\rangle \end{pmatrix} \begin{pmatrix} \langle R|x\rangle & \langle R|y\rangle \\ \langle L|x\rangle & \langle L|y\rangle \end{pmatrix} \\ & = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix} \begin{pmatrix} \hbar & 0 \\ 0 & -\hbar \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix} = \hbar \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \end{aligned}$$

In this basis, the matrix has only off-diagonal elements. Since a Hermitian matrix is equal to its transpose, complex conjugate, both of these representations for $\hat{J}_z$ satisfy this condition, as they must.

2.8 Summary:

rotations, bases, the identity operator.

"When we do not make a measurement that permits us to distinguish the intermediate states $|a_1\rangle$ and $|a_2\rangle$, we add the amplitudes and then square to get the probability, while if

we do make a measurement that can distinguish which of the states $|a_1\rangle$ and $|a_2\rangle$ the particle is in, we add the individual probabilities, not the amplitudes.